

To: The Editors-in-Chief, *Quantum Information Processing*

Re: Submission of an original research article

Dear Professors Pathak and Weinstein,

We are pleased to submit our manuscript, “**How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection**,” for consideration as an original article in *Quantum Information Processing*.

The question. Quantum kernel methods are among the most widely deployed quantum machine learning algorithms, and network intrusion detection is one of the application domains in which near-perfect accuracy is most often reported for them. Whether such results reflect the quantum feature map, or the classical processing that surrounds it, is a question of quantum information science rather than of security: it asks when a quantum kernel provides an advantage that a classical kernel cannot. We answer it for a realistic learning task, with a reproducible protocol and a result we did not expect.

What we did. We benchmark hybrid variational circuits and quantum-kernel SVMs against five tuned classical baselines on four intrusion-detection datasets under one leakage-controlled protocol, with a quantum-attribution audit of parameter-matched classical controls. The audit’s strongest positive was a quantum kernel that out-ranked its classical surrogate, and it survived both false-discovery-rate and family-wise Holm correction. We then dissected that result at the level of the kernels themselves, placing quantum and classical kernels on identical data with exact Gram matrices and equal tuning budgets, and applying the geometric difference of Huang et al. as a label-free test of whether a classical kernel can reproduce the quantum one.

What we found. The certified-significant advantage was produced by the classical baseline. Its surrogate differed from the quantum kernel in four ways besides the feature map, and once those are removed the gap is set entirely by the bandwidth grid searched for the classical kernel: a narrow grid yields an apparent quantum ROC-AUC advantage of 0.167, a wide grid leaves 0.007, with the classical kernel ahead at the 1% false-positive operating point. The mechanism is a distribution shift, since the test set contains 17 attack types absent from training, that in-distribution cross-validation cannot see; on a second dataset without such a shift, standard selection is reliable and classical kernels win outright. The geometric difference places the IQP quantum kernel within 1.4 to 2.4 of a classical kernel built on the circuit’s own phases on all four datasets, and that classical kernel reproduces it with no quantum state. Inducing the shift on the second dataset, by holding whole attack families out of its training data, recreates the failure on demand; leave-attack-types-out cross-validation repairs selection on NSL-KDD but not under that induced shift, so we recommend reporting kernel comparisons under several selection rules with the oracle bound. The projected quantum kernel is robust to finite-shot measurement down to 16 shots per basis. Finally we execute the projected kernel on three IBM Heron processors, where its measured Gram matrix correlates at 0.97 to 0.98 with the exact one and the classifier built on measured features stays within 0.01 ROC-AUC of the exact kernel, and we run the quantum layer of the four-qubit hybrid, the audit’s one surviving positive, on the same three devices, where its low-false-positive margin over a tuned Random Forest reproduces.

Why this journal. The contribution is a method for adjudicating claims of quantum-kernel advantage, demonstrated on a task where such claims are frequent. It builds on work that has appeared in this journal on the comparison of quantum and classical kernel functions in support vector machines (Abdulsalam and Ahmad, 2025) and on the training of quantum kernels (Innan et al., 2023), and on the broader scrutiny of quantum kernels by Schnabel and Roth, Flórez-Ablan et al. and Huang et al. Its central lesson, that statistical significance is not attribution and that a distribution shift can let the classical baseline’s tuning manufacture or erase an

apparent quantum advantage, applies to any quantum learning benchmark evaluated on shifted data.

Limitations we state up front. The variational results and the noise sweep are simulated at eight qubits; only the projected kernel and the hybrid’s quantum layer were run on hardware, as inference with 700 and 2,000 samples per device and default readout mitigation. The seed budget is small and uneven, and we report it exactly. Shift-aware selection requires attack-type labels and can only anticipate novelty of a kind the training data contains, which the induced shift makes explicit.

Reproducibility. The code, configurations, fixed seeds, the scripts that reconstruct our exact splits, the kernel-level analysis, the raw device measurements, and the per-run result records are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>; every table in the kernel analysis is generated directly from the result files. An earlier version of the benchmark, without the kernel-level analysis, is available as a preprint (arXiv:2608.18155); the kernel-level analysis is new to this submission.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submission and declare no competing interests.

Thank you for your consideration.

Sincerely,

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